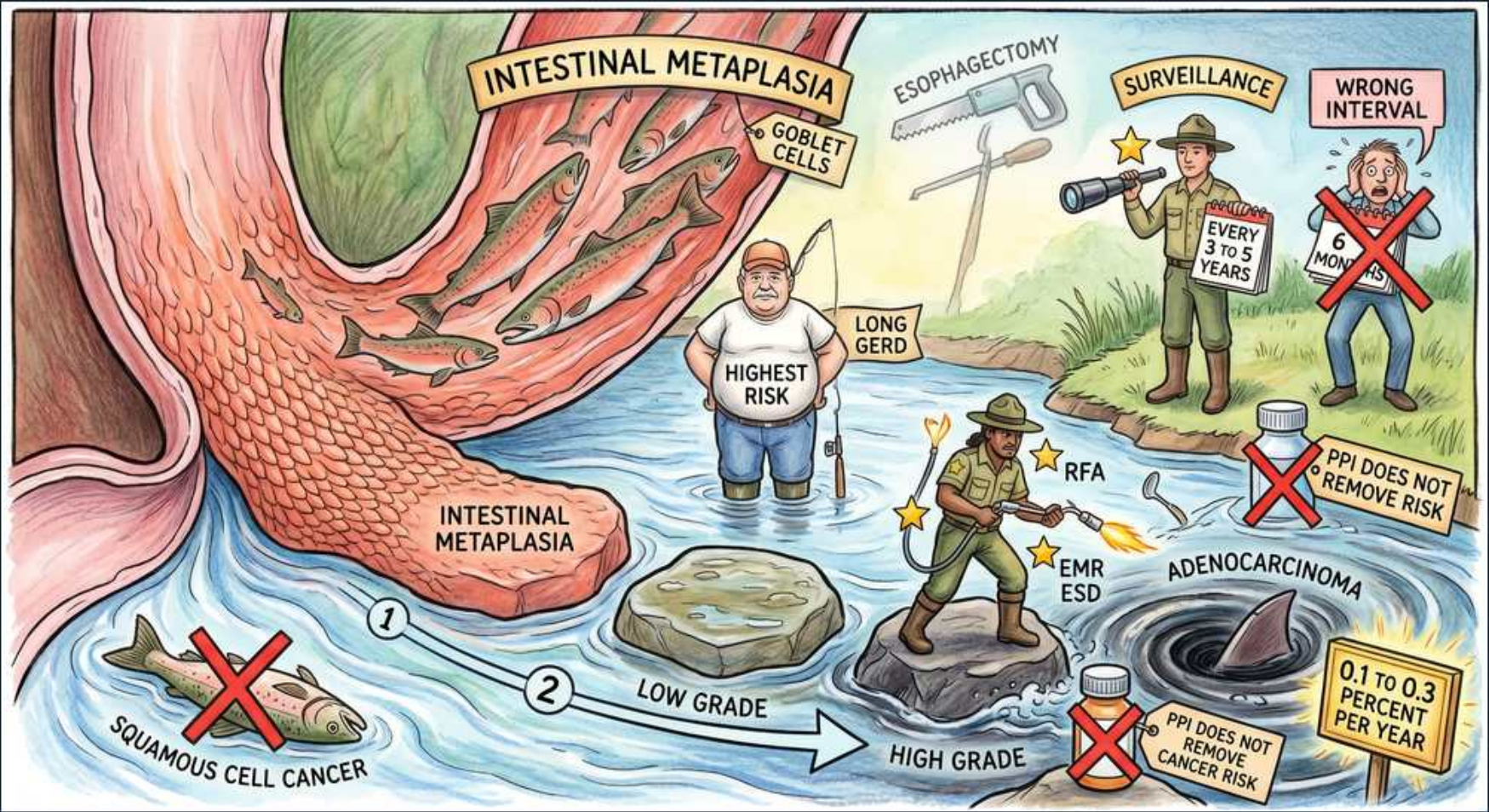


Esophageal — Barrett's Esophagus



1. Intestinal metaplasia

WHAT YOU SEE

Banner 'INTESTINAL METAPLASIA' over the salmon-pink fish-scale esophageal lining.

WHAT IT ENCODES

Barrett's esophagus = the normal squamous epithelium of the distal esophagus is replaced by intestinal-type columnar epithelium (metaplasia), an adaptive response to chronic acid. It extends upward from the GE junction (salmon-colored on endoscopy).

2. Goblet cells

WHAT YOU SEE

Star pointing to 'GOBLET CELLS' label on the scaly lining.

WHAT IT ENCODES

Histologic hallmark required for diagnosis: specialized intestinal metaplasia with GOBLET cells (mucin-containing, stain with Alcian blue). Columnar lining without goblet cells does not qualify as Barrett's in most definitions.

3. Risk: chronic GERD, obese white male

WHAT YOU SEE

Fisherman holding a 'HIGHEST RISK' sign with 'LONG GERD' label.

WHAT IT ENCODES

Risk factors: long-standing GERD, age >50, male sex, white race, central obesity, smoking, hiatal hernia. The classic stem patient is an older obese white man with years of reflux. These same factors raise esophageal adenocarcinoma risk.

4. Progression metaplasia->dysplasia->cancer

WHAT YOU SEE

Arrow path '1 -> 2 LOW GRADE -> HIGH GRADE' across the stream.

WHAT IT ENCODES

Stepwise progression: intestinal metaplasia -> low-grade dysplasia -> high-grade dysplasia -> invasive adenocarcinoma. Dysplasia grade drives both surveillance interval and treatment. Each step raises cancer risk.

5. Adenocarcinoma ~0.1-0.3%/yr

WHAT YOU SEE

Shark fin in a whirlpool labeled 'ADENOCARCINOMA' with a card '0.1 TO 0.3 PERCENT PER YEAR'.

WHAT IT ENCODES

Non-dysplastic Barrett's progresses to esophageal ADENOCarcinoma at ~0.1-0.3% per year. Barrett's is the main precursor lesion for distal esophageal adenocarcinoma. Cancer risk rises sharply with high-grade dysplasia.

6. Squamous cell cancer (RED X)

WHAT YOU SEE

Fish with a red X labeled 'SQUAMOUS CELL CANCER'.

WHAT IT ENCODES

Barrett's leads to ADENOCarcinoma, NOT squamous cell carcinoma. Squamous cell esophageal cancer is linked to smoking + alcohol and is mid/upper esophagus; it is not the Barrett's pathway.

BOARD TRAP

'Barrett's progresses to squamous cell carcinoma' -> wrong; it leads to adenocarcinoma; SCC is the smoking/alcohol upper-esophagus cancer.

7. Surveillance every 3-5 yr

WHAT YOU SEE

Ranger with a telescope, gold star, sign 'EVERY 3 TO 5 YEARS', banner 'SURVEILLANCE'.

WHAT IT ENCODES

Non-dysplastic Barrett's: surveillance endoscopy with 4-quadrant biopsies every 3-5 years. Low-grade dysplasia -> ablation or scope every 6-12 months; high-grade dysplasia -> endoscopic eradication. Continue PPI to control reflux.

8. Wrong interval 6 months (RED X)

WHAT YOU SEE

Alarmed figure with red X 'WRONG INTERVAL' and a '6 MONTHS' clock.

WHAT IT ENCODES

Re-scoping plain non-dysplastic metaplasia every 6 months is wrong/over-surveillance; the correct interval is 3-5 years. The 6-12 month interval applies only once low-grade dysplasia is found.

BOARD TRAP

'surveillance every 6 months for non-dysplastic Barrett's' -> wrong; correct interval is 3-5 years.

9. High-grade dysplasia -> RFA/EMR/ESD (gold star)

WHAT YOU SEE

Ranger with a flame torch and gold stars labeled 'RFA / EMR / ESD'.

WHAT IT ENCODES

High-grade dysplasia (and intramucosal cancer) is treated with endoscopic eradication therapy: EMR/ESD to resect visible/nodular lesions plus radiofrequency ablation (RFA) of the remaining flat Barrett's segment. This has largely replaced prophylactic esophagectomy.

10. Esophagectomy (historical)

WHAT YOU SEE

Saw labeled 'ESOPHAGECTOMY' in the background.

WHAT IT ENCODES

Esophagectomy was historically used for high-grade dysplasia but carries high morbidity/mortality; endoscopic eradication (RFA + EMR/ESD) is now preferred for HGD and intramucosal cancer. Reserve surgery for invasive disease beyond the mucosa.

11. PPI does not remove cancer risk (RED X)

WHAT YOU SEE

PPI bottle with a red X 'PPI DOES NOT REMOVE RISK' (upper).

WHAT IT ENCODES

PPIs control reflux symptoms and may slow progression but do NOT eliminate the cancer risk of established Barrett's; surveillance must continue and dysplasia still needs ablation/resection. PPI is adjunct, not cure.

BOARD TRAP

'PPI eliminates the cancer risk in Barrett's' -> wrong; it controls reflux but doesn't remove risk; surveillance continues.

12. PPI does not remove cancer risk (RED X, lower)

WHAT YOU SEE

Second PPI bottle with red X 'PPI DOES NOT REMOVE CANCER RISK' near high-grade.

WHAT IT ENCODES

Reinforced near the high-grade/cancer end: even with high-grade dysplasia, PPI alone is insufficient; definitive endoscopic eradication therapy is required. Acid suppression does not reverse dysplasia.

BOARD TRAP

'PPI alone manages high-grade dysplasia' -> wrong; HGD needs endoscopic eradication (RFA + EMR/ESD), not just PPI.
